

AutomaticValues

March 11, 2025

1 Discussion: How diffusivities D and Henry-like coefficients k are automatically set in SFPPy

1.1 Preamble

Diffusivities $D_{i,k}$ and $k_{i,k}$ are properties that depend on the pair substance (indexed i) and on the medium where sorption/diffusion/desorption occurs (indexed k). For diffusivities, the medium is limited to food layers indexed in our publications from 1 to n (in SFPPy for `layers` indexed from 0 to `n-1`). For Henry-like coefficients, the index range of k includes also the food/food simulant indexed 0 (in SFPPy denoted `foodlayer`).

1.2 Update the SFPPy path

```
[3]: import sys

# Check if SFPPy path is already in sys.path
SFPPy_path = '/home/olivi/natacha/python'
if SFPPy_path in sys.path:
    print(f"SFPPy installation '{SFPPy_path}' is already in sys.path.")
else:
    print(f"Add SFPPy path '{SFPPy_path}' to sys.path.")
    sys.path.append(SFPPy_path)
```

Add SFPPy path '/home/olivi/natacha/python' to sys.path.

1.3 Let's define four migrants: $i = 0..3$ (Python indexing)

```
[5]: from patankar.loadpubchem import migrant
list_of_migrants = ["toluene", "limonene", "BHT", "Irganox 1076"]
m = [migrant(s) for s in list_of_migrants]
print(m)
```

```
<migrant object>
Compound: toluene
  Name: ['phenyl-methane', ' [...] for HPLC, >=99.9%']
  cid: 1140
  CAS: ['25013-04-1', '108-88-3']
  M (min): 92.14
  M_array: [92.14]
```

```

    formula: C7H8
    smiles: CC1=CC=CC=C1
    InChiKey: YXFVVABEGXRONW-UHFFFAOYSA-N
    logP: [2.7]
    P' (calc): [1.23097093]
<migrant object>
    Compound: limonene
    Name: ['1-Methyl-4-isoprop [...] pentene, p.a., 95%']
    cid: 22311
    CAS: ['138-86-3', '7705-14-8']
    M (min): 136.23
    M_array: [136.23]
    formula: C10H16
    smiles: CC1=CCC(CC1)C(=C)C
    InChiKey: XMGQYMWWDOXHJM-UHFFFAOYSA-N
    logP: [3.4]
    P' (calc): [0.]
<migrant object>
    Compound: BHT
    Name: ['2,6-di-t butyl-4-m [...] ', 'Antioxidant 29']
    cid: 31404
    CAS: ['128-37-0']
    M (min): 220.35
    M_array: [220.35]
    formula: C15H24O
    smiles: CC1=CC(=C(C(=C1)C(C)(C)C)O)C(C)(C)C
    InChiKey: NLZUEZXRPGMBCV-UHFFFAOYSA-N
    logP: [5.3]
    P' (calc): [0.]
<migrant object>
    Compound: Irganox 1076
    Name: ['Octadecyl 3(3,5dit [...] e', 'Tominokusu SS']
    cid: 16386
    CAS: ['2082-79-3']
    M (min): 530.9
    M_array: [530.9]
    formula: C35H62O3
    smiles: CCCCCCCCCCCCCCCCCOC [...] )C(C)(C)C)O)C(C)(C)C
    InChiKey: SSDSCDGV MJFTEQ-UHFFFAOYSA-N
    logP: [13.8]
    P' (calc): [0.]
[<migrant: phenyl-methane - M=92.14 g/mol>, <migrant: 1-Methyl [...] lohexene -
M=136.23 g/mol>, <migrant: 2,6-di-t [...] ylphenol - M=220.35 g/mol>, <migrant:
Octadecyl [...] opionate - M=530.9 g/mol>]

```

1.4 Let's define four polymers and a temperature $T = 50^{\circ}\text{C}$

- PET with two states: dry PET (gPET for glassy PET) and plasticized PET (wPET for *wet* PET)
- PS (general purpose PS, unplasticized)
- PP (atactic polypropylene)

```
[7]: from patankar.layer import gPET, wPET, PS, PP
materials = gPET()+wPET()+PS()+PP()
```

1.5 Now we incorporate $i = 0$ in all layers materials

Without inserting a substance in multilayer, all layers have the same diffusivity: D:
1e-14 [m**2/s]

The default $k = 1$ value is also assigned by default to all layers.

```
[10]: print("D=",materials.D) # we print the default values assigned
materials
```

```
D= [1.e-14 1.e-14 1.e-14 1.e-14]
```

```
[LAYER object version=1, contact=olivier.vitrac@agroparistech.fr]
```

```
4-multilayer of LAYER object:
```

```
-- [ layer 1 of 4 ] ----- barrier rank=2 -----
    name: "layer in gPET"
    type: "polymer"
    material: "glassy PET"
    code: "PET"
        l: 0.0002 [m]
        D: 1e-14 [m**2/s]
        k: 1 [a.u.]
        C0: 1000 [a.u.]
-- [ layer 2 of 4 ] ----- barrier rank=4 -----
    name: "layer in wPET"
    type: "polymer"
    material: "plasticized PET"
    code: "PET"
        l: 0.0002 [m]
        D: 1e-14 [m**2/s]
        k: 1 [a.u.]
        C0: 1000 [a.u.]
-- [ layer 3 of 4 ] ----- barrier rank=3 -----
    name: "layer in PS"
    type: "polymer"
    material: "polystyrene"
    code: "PS"
        l: 0.0001 [m]
        D: 1e-14 [m**2/s]
```

```

        k: 1 [a.u.]
        C0: 1000 [a.u.]
-- [ layer 4 of 4 ] ----- barrier rank=1 -----
    name: "layer in PP"
    type: "polymer"
    material: "isotactic polypropylene"
    code: "PP"
        l: 0.0003 [m]
        D: 1e-14 [m**2/s]
        k: 1 [a.u.]
        C0: 1000 [a.u.]

```

[10]: <multilayer with 4 layers: ['layer in gPET', 'layer in wPET', 'layer in PS', 'layer in PP']>

1.5.1 We insert $m[0]$ in all layers

All values of D and k are updated automatically;

```

[12]: materials.update(migrant=m[0],T=50)
print(materials.D) # we print the default values automatically calculated at
↪temperature T
materials

```

dimension mismatch, the last value has been repeated
[1.86299457e-17 1.24073529e-15 2.64579177e-20 2.78163880e-12]

[LAYER object version=1, contact=olivier.vitrac@agroparistech.fr]

4-multilayer of LAYER object:

```

-- [ layer 1 of 4 ] ----- barrier rank=1 -----
    name: "layer in gPET"
    type: "polymer"
    material: "glassy PET"
    code: "PET"
    crystal: 0.35
        Tg: 76 [degC]
        l: 0.0002 [m]
        D: 1.863e-17 [m**2/s]
        = DFV(gPET,<migrant: phenyl-methane - M=92.14 g/mol>,T=50.0 degC)
        k: 0.565 [a.u.]
        = kFHP(<ethylene terephthalate>,<migrant: phenyl-methane - M=92.14
g/mol>)
        C0: 1000 [a.u.]
        T: 50 [degC]
-- [ layer 2 of 4 ] ----- barrier rank=3 -----
    name: "layer in wPET"
    type: "polymer"
    material: "plasticized PET"
    code: "PET"

```

```

crystal: 0.35
  Tg: 46 [degC]
  l: 0.0002 [m]
  D: 1.241e-15 [m**2/s]
    = DFV(wPET,<migrant: phenyl-methane - M=92.14 g/mol>,T=50.0 degC)
  k: 0.565 [a.u.]
    = kFHP(<ethylene terephthalate>,<migrant: phenyl-methane - M=92.14
g/mol>)
  C0: 1000 [a.u.]
  T: 50 [degC]
-- [ layer 3 of 4 ] ----- barrier rank=4 -----
  name: "layer in PS"
  type: "polymer"
material: "polystyrene"
  code: "PS"
  crystal: 0
  Tg: 100 [degC]
  l: 0.0001 [m]
  D: 2.646e-20 [m**2/s]
    = DFV(PS,<migrant: phenyl-methane - M=92.14 g/mol>,T=50.0 degC)
  k: 0.3672 [a.u.]
    = kFHP(<styrene>,<migrant: phenyl-methane - M=92.14 g/mol>)
  C0: 1000 [a.u.]
  T: 50 [degC]
-- [ layer 4 of 4 ] ----- barrier rank=2 -----
  name: "layer in PP"
  type: "polymer"
material: "isotactic polypropylene"
  code: "PP"
  crystal: 0.5
  Tg: 0 [degC]
  l: 0.0003 [m]
  D: 2.782e-12 [m**2/s]
    = Dpiringer(PP,<migrant: phenyl-methane - M=92.14 g/mol>,T=50.0 degC)
  k: 1.106 [a.u.]
    = kFHP(<propylene>,<migrant: phenyl-methane - M=92.14 g/mol>)
  C0: 1000 [a.u.]
  T: 50 [degC]
[12]: <multilayer with 4 layers: ['layer in gPET', 'layer in wPET', 'layer in PS',
'layer in PP']>

```

1.5.2 Interpretation

We note that the diffusivities are different for all four layers: $>[1.86299457\text{e-}17$
 $1.24073529\text{e-}15$ $2.64579177\text{e-}20$ $2.78163880\text{e-}12]$

The output is giving details on the models, which were used | Reported model | Model |
Interpretation | | |

```

| | DFV(gPET,<migrant: Toluene-13C - M=92.14
g/mol>,T=50.0 degC) | DFV | best model for toluene when Free-volume parame-
ters are available for this polymer | | DFV(wPET,<migrant: Toluene-13C - M=92.14
g/mol>,T=50.0 degC) | DFV | best model for toluene when Free-volume parame-
ters are available for this polymer | | DFV(gPS,<migrant: Toluene-13C - M=92.14
g/mol>,T=50.0 degC) | DFV | best model for toluene when Free-volume parameters
are available for this polymer | | Dpiringer(PP,<migrant: Toluene-13C - M=92.14
g/mol>,T=50.0 degC) | Dpiringer | fall-back mechanism |

```

1.5.3 We insert m[1] in all layers

```

[15]: materials.update(migrant=m[1],T=50)
print(materials.D) # we print the default values automatically calculated at
↳ temperature T
materials

```

dimension mismatch, the last value has been repeated
[2.70631048e-19 1.72090422e-15 3.74480502e-21 1.39807262e-12]

[LAYER object version=1, contact=olivier.vitrac@agroparistech.fr]

4-multilayer of LAYER object:

```

-- [ layer 1 of 4 ] ----- barrier rank=1 -----
    name: "layer in gPET"
    type: "polymer"
    material: "glassy PET"
    code: "PET"
    crystal: 0.35
    Tg: 76 [degC]
    l: 0.0002 [m]
    D: 2.706e-19 [m**2/s]
    = Dwelle(gPET,<migrant: 1-Methyl [...] lohene - M=136.23
g/mol>,T=50.0 degC)
    k: 0.8452 [a.u.]
    = kFHP(<ethylene terephthalate>,<migrant: 1-Methyl [...] lohene -
M=136.23 g/mol>)
    CO: 1000 [a.u.]
    T: 50 [degC]
-- [ layer 2 of 4 ] ----- barrier rank=3 -----
    name: "layer in wPET"
    type: "polymer"
    material: "plasticized PET"
    code: "PET"
    crystal: 0.35
    Tg: 46 [degC]
    l: 0.0002 [m]
    D: 1.721e-15 [m**2/s]
    = Dpiringer(wPET,<migrant: 1-Methyl [...] lohene - M=136.23
g/mol>,T=50.0 degC)

```

```

        k: 0.8452 [a.u.]
        = kFHP(<ethylene terephthalate>,<migrant: 1-Methyl [...] lohexene -
M=136.23 g/mol>)
        C0: 1000 [a.u.]
        T: 50 [degC]
-- [ layer 3 of 4 ] ----- barrier rank=4 -----
        name: "layer in PS"
        type: "polymer"
        material: "polystyrene"
        code: "PS"
        crystal: 0
        Tg: 100 [degC]
        l: 0.0001 [m]
        D: 3.745e-21 [m**2/s]
        = Dwelle(PS,<migrant: 1-Methyl [...] lohexene - M=136.23 g/mol>,T=50.0
degC)
        k: 0.5494 [a.u.]
        = kFHP(<styrene>,<migrant: 1-Methyl [...] lohexene - M=136.23 g/mol>)
        C0: 1000 [a.u.]
        T: 50 [degC]
-- [ layer 4 of 4 ] ----- barrier rank=2 -----
        name: "layer in PP"
        type: "polymer"
        material: "isotactic polypropylene"
        code: "PP"
        crystal: 0.5
        Tg: 0 [degC]
        l: 0.0003 [m]
        D: 1.398e-12 [m**2/s]
        = Dpiringer(PP,<migrant: 1-Methyl [...] lohexene - M=136.23
g/mol>,T=50.0 degC)
        k: 4.322 [a.u.]
        = kFHP(<propylene>,<migrant: 1-Methyl [...] lohexene - M=136.23
g/mol>)
        C0: 1000 [a.u.]
        T: 50 [degC]
[15]: <multilayer with 4 layers: ['layer in gPET', 'layer in wPET', 'layer in PS',
'layer in PP']>

```

1.5.4 Interpretation

We note that the diffusivities are different for all four layers:

```
[2.70631048e-19 1.72090422e-15 3.74480502e-21 1.39807262e-12]
```

The output is giving details on the models, which were used

Reported model	Model	Interpretation
Dwelle(gPET,<migrant: Cyclohex [...])-, (R)- - M=136.23 g/mol>,T=50.0 degC)	Dwelle	best model for glassy PET when DFV is not available
Dpiringer(wPET,<migrant: Cyclohex [...])-, (R)- - M=136.23 g/mol>,T=50.0 degC)	Dpiringerfall-back mechanism	
Dwelle(PS,<migrant: Cyclohex [...])-, (R)- - M=136.23 g/mol>,T=50.0 degC)	Dwelle	best model for glassy PS when DFV is not available
Dpiringer(PP,<migrant: Cyclohex [...])-, (R)- - M=136.23 g/mol>,T=50.0 degC)	Dpiringerfall-back mechanism	

1.6 We can repeat the mechanism for m[2] and m[3]

```
[18]: print("D for",m[2].compound,"=",materials.update(migrant=m[2]).D)
      print("D for",m[3].compound,"=",materials.update(migrant=m[3]).D)
```

```
D for BHT = [2.13915138e-20 5.73730849e-16 8.90819931e-24 4.66102284e-13]
D for Irganox 1076 = [1.88965546e-23 2.87174640e-17 4.78703852e-31
2.33302351e-14]
```

```
[19]: materials.update(migrant=m[2])
```

```
[LAYER object version=1, contact=olivier.vitrac@agroparistech.fr]
4-multilayer of LAYER object:
-- [ layer 1 of 4 ] ----- barrier rank=1 -----
    name: "layer in gPET"
    type: "polymer"
    material: "glassy PET"
    code: "PET"
    crystal: 0.35
    Tg: 76 [degC]
    l: 0.0002 [m]
    D: 2.139e-20 [m**2/s]
      = Dwelle(gPET,<migrant: 2,6-di-t [...] ylphenol - M=220.35
g/mol>,T=50.0 degC)
    k: 1.387 [a.u.]
      = kFHP(<ethylene terephthalate>,<migrant: 2,6-di-t [...] ylphenol -
M=220.35 g/mol>)
    CO: 1000 [a.u.]
    T: 50 [degC]
-- [ layer 2 of 4 ] ----- barrier rank=3 -----
    name: "layer in wPET"
    type: "polymer"
```



```

material: "plasticized PET"
  code: "PET"
  crystal: 0.35
    Tg: 46 [degC]
    l: 0.0002 [m]
    D: 5.737e-16 [m**2/s]
      = Dpiringer(wPET,<migrant: 2,6-di-t [...] ylphenol - M=220.35
g/mol>,T=50.0 degC)
    k: 1.387 [a.u.]
      = kFHP(<ethylene terephthalate>,<migrant: 2,6-di-t [...] ylphenol -
M=220.35 g/mol>)
    C0: 1000 [a.u.]
    T: 50 [degC]
-- [ layer 3 of 4 ] ----- barrier rank=4 -----
  name: "layer in PS"
  type: "polymer"
  material: "polystyrene"
  code: "PS"
  crystal: 0
    Tg: 100 [degC]
    l: 0.0001 [m]
    D: 8.908e-24 [m**2/s]
      = Dwelle(PS,<migrant: 2,6-di-t [...] ylphenol - M=220.35 g/mol>,T=50.0
degC)
    k: 0.9015 [a.u.]
      = kFHP(<styrene>,<migrant: 2,6-di-t [...] ylphenol - M=220.35 g/mol>)
    C0: 1000 [a.u.]
    T: 50 [degC]
-- [ layer 4 of 4 ] ----- barrier rank=2 -----
  name: "layer in PP"
  type: "polymer"
  material: "isotactic polypropylene"
  code: "PP"
  crystal: 0.5
    Tg: 0 [degC]
    l: 0.0003 [m]
    D: 4.661e-13 [m**2/s]
      = Dpiringer(PP,<migrant: 2,6-di-t [...] ylphenol - M=220.35
g/mol>,T=50.0 degC)
    k: 7.093 [a.u.]
      = kFHP(<propylene>,<migrant: 2,6-di-t [...] ylphenol - M=220.35
g/mol>)
    C0: 1000 [a.u.]
    T: 50 [degC]

```

```

[19]: <multilayer with 4 layers: ['layer in gPET', 'layer in wPET', 'layer in PS',
'layer in PP']>

```

```
[20]: materials.update(migrant=m[3])
```

```
[LAYER object version=1, contact=olivier.vitrac@agroparistech.fr]
4-multilayer of LAYER object:
-- [ layer 1 of 4 ] ----- barrier rank=1 -----
    name: "layer in gPET"
    type: "polymer"
    material: "glassy PET"
    code: "PET"
    crystal: 0.35
    Tg: 76 [degC]
    l: 0.0002 [m]
    D: 1.89e-23 [m**2/s]
    = Dwelle(gPET,<migrant: Octadecy [...] opionate - M=530.9
g/mol>,T=50.0 degC)
    k: 3.431 [a.u.]
    = kFHP(<ethylene terephthalate>,<migrant: Octadecy [...] opionate -
M=530.9 g/mol>)
    C0: 1000 [a.u.]
    T: 50 [degC]
-- [ layer 2 of 4 ] ----- barrier rank=3 -----
    name: "layer in wPET"
    type: "polymer"
    material: "plasticized PET"
    code: "PET"
    crystal: 0.35
    Tg: 46 [degC]
    l: 0.0002 [m]
    D: 2.872e-17 [m**2/s]
    = Dpiringer(wPET,<migrant: Octadecy [...] opionate - M=530.9
g/mol>,T=50.0 degC)
    k: 3.431 [a.u.]
    = kFHP(<ethylene terephthalate>,<migrant: Octadecy [...] opionate -
M=530.9 g/mol>)
    C0: 1000 [a.u.]
    T: 50 [degC]
-- [ layer 3 of 4 ] ----- barrier rank=4 -----
    name: "layer in PS"
    type: "polymer"
    material: "polystyrene"
    code: "PS"
    crystal: 0
    Tg: 100 [degC]
    l: 0.0001 [m]
    D: 4.787e-31 [m**2/s]
    = Dwelle(PS,<migrant: Octadecy [...] opionate - M=530.9 g/mol>,T=50.0
degC)
```

```

        k: 2.23 [a.u.]
        = kFHP(<styrene>,<migrant: Octadecy [...] opionate - M=530.9 g/mol>)
    CO: 1000 [a.u.]
    T: 50 [degC]
-- [ layer 4 of 4 ] ----- barrier rank=2 -----
    name: "layer in PP"
    type: "polymer"
    material: "isotactic polypropylene"
    code: "PP"
    crystal: 0.5
    Tg: 0 [degC]
    l: 0.0003 [m]
    D: 2.333e-14 [m**2/s]
    = Dpiringer(PP,<migrant: Octadecy [...] opionate - M=530.9
g/mol>,T=50.0 degC)
    k: 17.55 [a.u.]
    = kFHP(<propylene>,<migrant: Octadecy [...] opionate - M=530.9 g/mol>)
    CO: 1000 [a.u.]
    T: 50 [degC]
[20]: <multilayer with 4 layers: ['layer in gPET', 'layer in wPET', 'layer in PS',
'layer in PP']>

```

1.7 Adapt automatically k value for food

We choose here ethanol as food simulant

```

[45]: from patankar.food import ethanol
      contactmedium = ethanol(T=50)
      repr(contactmedium)

```

Food object "ethanol" (ethanol = from pure ethanol down to ethanol 95%) with properties:

```

      CF0: 0 [a.u.]
contacttime: 864000 [s]
      density: 1000 [kilogram / meter ** 3]
      h: 0.0001 [meter / second]
surfacearea: 0.06 [meter ** 2]
      volume: 0.001 [meter ** 3]

```

```

[45]: '<ethanol: ethanol>'

```

1.7.1 Add $m[0]$ to contactmedium

This step makes appear the field k_0 > k_0 : 0.17613565 [a.u.] > > =
kFHP(<ethanol>,<migrant: phenyl-methane - M=92.14 g/mol>)

```

[58]: contactmedium.update(migrant=m[0])

```

Food object "ethanol" (ethanol = from pure ethanol down to ethanol 95%) with properties:

```
CF0: 0 [a.u.]
contacttime: 864000 [s]
density: 1000 [kilogram / meter ** 3]
h: 0.0001 [meter / second]
k0: 0.17613565 [a.u.]
    = kFHP(<ethanol>,<migrant: phenyl-methane - M=92.14 g/mol>)
surfacearea: 0.06 [meter ** 2]
volume: 0.001 [meter ** 3]
```

```
[58]: <ethanol: ethanol>
```

```
print('k0 for',m[0].compound," = ",contactmedium.k0)
```

```
[61]: print('k0 for ',m[0].compound," = ",contactmedium.k0)
```

```
k0 for toluene = [0.17613565]
```

1.7.2 We repeat the process for m[1], m[2] and m[3]

```
[66]: print('k0 for ',m[1].compound," = ",contactmedium.update(migrant=m[1]).k0)
      print('k0 for ',m[2].compound," = ",contactmedium.update(migrant=m[2]).k0)
      print('k0 for ',m[3].compound," = ",contactmedium.update(migrant=m[3]).k0)
```

```
k0 for limonene = [0.55908817]
```

```
k0 for BHT = [0.25828644]
```

```
k0 for Irganox 1076 = [0.63893733]
```